Sanyam Rana

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Education

(B.Tech) Indian Institute of Technology Jodhpur 6.41/10
Bachelor of Technology – Electrical Engineering — Aug '24 – May '28
Courses: Data Structures and Algorithms, Pattern Recognition in Machine Learning (3+ core systems courses)

(12th) R.K.Educational School, Sikandrabad -'24 89% (Aggregate)

(10th) St.Mary's Academy, Meerut -'22 95% (Aggregate)

Projects

SPAC3D: AI-Powered Interior Intelligence & Recommendation System *GitHub — Apr '26*
Developed a computer vision pipeline using YOLO-World for open-vocabulary object detection, enabling zero-shot localization of interior elements from indoor scene images through deep learning-based visual feature extraction and semantic understanding.
Tech Stack: Python, PyTorch, YOLO-World, ResNet50, Scikit-learn, FastAPI, React, TypeScript, Docker, NumPy

- Built an interior style classification framework leveraging **ResNet50 feature extraction**, **PCA dimensionality reduction**, and **k-Nearest Neighbors (kNN)** and **SVM** classifiers to categorize rooms into six design styles.
- Implemented custom post-processing algorithms including \textbf{Intersection-over-Area (IoA) suppression} to improve object detection quality, enhance prediction accuracy, and reduce redundant predictions for downstream inference tasks.
- Designed a **content-based recommendation system** using **vector embeddings** and **cosine similarity** search to retrieve aesthetically relevant furniture products, leveraging semantic feature matching information retrieval techniques.

EV Route Optimization System *GitHub — Website, Nov '25*
Architected a graph-based EV routing engine using Dijkstra's and A* algorithms across 10^4+ nodes per query, enabling scalable path optimization for transportation networks with efficient decision making while reducing energy consumption by $\sim$ 25%.
Tech Stack: C++, Dijkstra's Algorithm, A Algorithm, OpenStreetMap, React, Cytoscape.js*

- Implemented Dijkstra's & A* algorithms for route optimization across 100+ networks, saving $\sim$ 25% Energy
- Formulated elevation-aware cost functions using SOC constraints, reducing required charging stops by $\sim$ 25% and improving route feasibility under real-world driving conditions for long-distance electric vehicle transportation scenarios.
- Engineered a modular backend API and real-time map UI handling 1K+ nodes per query, improving system maintainability by $\sim$ 40% and enabling scalable graph traversal for dynamic routing across complex transportation network simulations.

Multi-Agent Trading & Financial Prediction System *GitHub, Jan '25*
Architected a cooperative multi-agent financial engine and machine learning pipeline to process real-time market data, forecast asset trajectories, and compile automated stock intelligence reports.
Tech Stack: Python, FastAPI, React, Redux Toolkit, Charts, Scikit-learn, Redis, OpenAI API, Gemini API, Uvicorn, Pandas

- Engineered a multi-agent cooperative routing architecture coordinating 5 specialized agents (Technical, Sentiment, Fundamental, Risk, Portfolio) to parallelize complex financial query analysis, reducing response latency by 35%.
- Implemented an automated machine learning forecasting pipeline using Random Forest models to train on technical indicators, predicting short-term asset direction and target prices with 76% accuracy across market trading conditions .
- Developed a high-performance interactive dashboard using Lightweight Charts and Tailwind CSS, rendering custom candlestick/volume series and dynamic overlays for RSI, EMA, and Bollinger Bands with visualization capabilities.
- Formulated an asynchronous caching layer using Redis and automated mock data fallbacks, preventing external API rate-limiting issues and reducing provider dependency by \sim50% during high-frequency financial data retrieval operations.

Technical Skills

- **Languages:** Python (2+ yrs), C/C++ (2+ yrs), JavaScript
- **Machine Learning:** NumPy, Pandas, Scikit-learn, OpenCV, NLP, Embeddings, Recommendation Systems
- **Web / Frameworks:** React, FastAPI, HTML, CSS, REST APIs
- **Databases:** SQLite, Firebase
- **Developer Tools:** Git, GitHub, VS Code, Docker
- **Embedded Systems:** Arduino Nano, Serial Communication, Motor Drivers

Extra Curriculars

- **Project Lead & Mentor – Managing a cryptography project involving threshold-based secret encoding and secure data recovery using Shamir's Secret Sharing principles** *IITJ, Present*
- **Assistant Head – Prometeo, led coordination of 10+ technical events with 1000+ participants** *IITJ, Jan '26*
- **Industry Project Mentee – *FinSage AI*, contributed to a multi-agent financial trading system in collaboration under WARP Project Series, with mentorship from Bhaskarjit Sarmah (Director, BlackRock)** *IITJ, Jan '25*
- **Volunteer – Prometeo, Ignus, Varchas, Supported 15+ inter-college events and workshops.** *IITJ, '25*